

Probability, Distributions, and Statistical Inference

INBRE Summer Biostatistics Workshop – Day 5

Alexander McLain

Summer 2026

What Is Probability?

- Probability is a number between 0 and 1 that measures how likely an event is.
 - $\Pr(\text{event}) = 0$ means it cannot happen.
 - $\Pr(\text{event}) = 1$ means it will certainly happen.
- A useful way to think about it: if we repeated an experiment many times, what fraction of the time would this event occur?
- **Example:** If 10% of a population carries a particular gene variant, then for a randomly selected individual, $\Pr(\text{carries variant}) = 0.10$.

Combining Probabilities

- **Union** (A or B): at least one of the two events occurs.

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

- **Intersection** (A and B): both events occur together.
- **Complement** (A^c): the event does *not* occur.

$$\Pr(A^c) = 1 - \Pr(A)$$

- **Example:** In a study sample, 47% are male and 50% are aged 18–44, with 20% being both male and aged 18–44.

$$\Pr(\text{male or aged 18–44}) = 0.47 + 0.50 - 0.20 = 0.77$$

Conditional Probability and Independence

- **Conditional probability** : the probability of A given that B has already occurred.

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

- **Example:** What is the probability that a patient tests positive for a disease, given they are in a high-risk group?
- Two events are **independent** if knowing one occurred tells you nothing about the other:

$$\Pr(A \mid B) = \Pr(A)$$

- **Example:** In many genetic studies, we assume that two SNPs on different chromosomes are inherited independently.

- A **random variable** is a quantity whose value is determined by a random process. We typically denote it with a capital letter, e.g., X .
- **Discrete** random variables take countable values $(0, 1, 2, \dots)$
 - Number of mutations in a DNA sequence
 - Number of patients who respond to treatment
- **Continuous** random variables can take any value in a range
 - Body weight, blood pressure, gene expression level
- A **probability distribution** describes all possible values and how likely each is.

Summarizing a Distribution: Mean and Standard Deviation

- The **mean** (μ) is the long-run average: the “center of mass” of the distribution.
- The **standard deviation** (σ) measures spread around the mean. A large σ means values are widely scattered; a small σ means they cluster tightly.
- From data, we estimate these with:
 - **Sample mean:** $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
 - **Sample standard deviation:** $s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$
- **Example:** Five crab weights (kg): 0.6, 0.7, 0.8, 0.9, 1.0. $\bar{X} = 0.8$ kg, $s \approx 0.16$ kg.

The Binomial Distribution

- Use the Binomial when you have n independent trials, each with the same probability p of “success.”
- $X \sim \text{Binomial}(n, p)$ counts the number of successes.
- Mean: $\mu = np$; Standard deviation: $\sigma = \sqrt{np(1 - p)}$
- **Biological examples:**
 - Number of patients (out of 20) who respond to a drug, if each has a 30% response rate
 - Number of offspring with a recessive trait in a cross
 - Number of individuals in a sample who carry a risk allele
- In : `rbinom(n, size, prob)`, `dbinom(x, size, prob)`, `pbinom(q, size, prob)`

The Poisson Distribution

- Use the Poisson for counts of rare events occurring over a fixed interval of time, area, or sequence length.
- $X \sim \text{Poisson}(\lambda)$, where λ is the average count.
- Mean: $\mu = \lambda$; Standard deviation: $\sigma = \sqrt{\lambda}$
- **Biological examples:**
 - Number of mutations per kilobase in a genome
 - Number of bacterial colonies on a plate
 - Number of rare disease cases in a county per year
 - Read counts in an RNA-seq experiment (approximately)
- In : `rpois(n, lambda)`, `dpois(x, lambda)`, `ppois(q, lambda)`

The Normal Distribution

- The Normal (or Gaussian) distribution is the classic “bell curve,” symmetric around its mean.
- Written $X \sim N(\mu, \sigma^2)$, characterized entirely by its mean μ and variance σ^2 .
- **A useful rule of thumb:**
 - $\approx 68\%$ of values fall within 1 standard deviation of the mean
 - $\approx 95\%$ fall within 2 standard deviations
 - $\approx 99.7\%$ fall within 3 standard deviations
- **Example:** Adult blue crab weights $\sim N(0.8 \text{ kg}, 0.3^2)$. A crab weighing more than 1.4 kg (i.e., $> \mu + 2\sigma$) is unusual.
- In : `rnorm(n, mean, sd)`, `pnorm(q, mean, sd)`, `qnorm(p, mean, sd)`

Normal Distribution: Worked Example

- Adult blue crab weights: $\mu = 0.8$ kg, $\sigma = 0.3$ kg.
- What is the probability a randomly selected crab weighs more than 1 kg?
- Step 1: How far is 1 kg from the mean, in standard deviation units?

$$z = \frac{1.0 - 0.8}{0.3} = 0.67$$

- Step 2: Look up (or compute in R): $\Pr(Z > 0.67) \approx 0.25$.
- **Interpretation:** About 25% of adult blue crabs weigh more than 1 kg.
- In : `1 - pnorm(1.0, mean = 0.8, sd = 0.3)`

The Central Limit Theorem (CLT)

- Suppose we measure the same quantity on n individuals and compute the sample mean $\bar{X}$.
- **Key fact:** Even if the individual measurements are not normally distributed, the *sample mean* $\bar{X}$ will be approximately normally distributed when n is large.
- Specifically:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

- As sample size grows, $\bar{X}$ becomes *more precise* (tighter distribution around μ).
- This is why the Normal distribution is so central to statistics: it describes averages, even when raw data are skewed or discrete.

- The **standard error** (SE) of the mean is $\sigma/\sqrt{n}$.
- The SE tells you how variable your estimate of the mean is across repeated studies.
- **Example:** Suppose gene expression measurements have $\sigma = 2.0$ (arbitrary units).
 - With $n = 4$ samples: $SE = 2.0/\sqrt{4} = 1.0$
 - With $n = 100$ samples: $SE = 2.0/\sqrt{100} = 0.2$
- Larger samples give more precise estimates of the mean – not because individuals vary less, but because averaging smooths out individual variability.
- **We will simulate this in the afternoon R lab.**

From Sample to Population

- In practice we never observe the whole population – we observe a *sample*.
- **Statistical inference** uses the sample to draw conclusions about the population.
- Two main tools:
 - **Hypothesis tests**: Is there evidence against a specific claim?
 - **Confidence intervals**: What is a plausible range for the true value?
- Both are built on the same foundation: how variable would our estimate be if we repeated the study many times?

Hypothesis Testing: The Core Idea

- Start with a **null hypothesis** (H_0): a default claim of “no effect” or “no difference.”
 - Example: “This drug does not change blood pressure compared to placebo.”
- Collect data. If the data would be very unlikely under H_0 , we have evidence against H_0 .
- **Analogy:** A jury assumes innocence until the evidence is overwhelming. We assume H_0 until the data compel us otherwise.
- The **p-value** quantifies “how unlikely.”

The p-value: What It Is

- The **p-value** is the probability of observing a result at least as extreme as the one we got, *assuming the null hypothesis is true*.
- A small p-value means: “If H_0 were true, this result would be very rare.” That is evidence against H_0 .
- Common threshold: $p < 0.05$ is called “statistically significant.”
- **Correct interpretation:** A p-value of 0.03 means there is a 3% chance of seeing data this extreme if H_0 is true.
- **Not** the probability that H_0 is true.
Not the probability your result is due to chance.
Not a measure of the size or importance of the effect.

The t-Distribution and Small Samples

- When we estimate σ from a small sample, there is extra uncertainty.
- The **t-distribution** accounts for this: it looks like a Normal distribution but has heavier tails (more probability of extreme values).
- As sample size grows, the t-distribution converges to the Normal.
- **When do we use it?** For testing means or computing confidence intervals when the population standard deviation is unknown – which is almost always.
- In : `t.test(x)` performs a one-sample t-test and returns the p-value, test statistic, and confidence interval automatically.

Confidence Intervals: What They Are

- A **confidence interval** (CI) gives a range of plausible values for the true population parameter.
- A 95% CI has the form:

$$\bar{X} \pm t^* \times \frac{s}{\sqrt{n}}$$

where t^* is a critical value from the t-distribution.

- **Correct interpretation:** If we repeated this study 100 times and built a 95% CI each time, about 95 of those intervals would contain the true mean.
- **Not** the probability that the true mean is in *this particular* interval (it either is or it isn't).
- Wider CI $\Rightarrow$ more uncertainty.
Narrower CI $\Rightarrow$ more precision (usually from larger n).

Statistical vs. Practical Significance

- **Statistical significance** ($p < 0.05$) tells you the effect is probably real, not due to chance.
- **Practical significance** tells you whether the effect is large enough to matter scientifically or clinically.
- **With a large enough sample, trivially small effects become statistically significant.**
- **Example:** A new drug lowers systolic blood pressure by 0.5 mmHg. In a study of $n = 100,000$ patients this may be highly significant ($p < 0.001$) yet clinically meaningless.
- Always report *effect sizes and confidence intervals*, not just p-values.

Common Tests We Will Use This Week

Situation	Test	R function
One group mean vs. a value	One-sample t-test	<code>t.test(x, mu = ...)</code>
Two group means	Two-sample t-test	<code>t.test(x, y)</code>
Two paired measurements	Paired t-test	<code>t.test(x, y, paired = TRUE)</code>
Count data, two groups	Chi-square test	<code>chisq.test(table)</code>
Non-normal, two groups	Wilcoxon rank-sum	<code>wilcox.test(x, y)</code>

- The t-tests assume approximately Normal data (or large n , by CLT).
- Chi-square tests are for categorical outcomes (counts, proportions).
- Wilcoxon tests make no distributional assumption – useful when data are skewed.

One-Sample t-Test: When to Use It

- **Question:** Is the mean of my group different from some known or hypothesized value?
- **Biological example:** A standard healthy fasting glucose level is 90 mg/dL. You measure glucose in 25 lab mice after a dietary intervention. Is their mean glucose different from 90?
- $H_0: \mu = \mu_0$ (the true mean equals the reference value)
- $H_A: \mu \neq \mu_0$ (two-sided), or $\mu > \mu_0$ / $\mu < \mu_0$ (one-sided)
- **What you need:** One continuous measurement per subject; data approximately Normal (or n large enough for the CLT to apply).

One-Sample t-Test: The Mechanics

- The test statistic is:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

where $\bar{X}$ is the sample mean, s is the sample SD, and μ_0 is the null value.

- Intuitively: how many standard errors is the sample mean away from the null value?
- Under H_0 , t follows a t-distribution with $n - 1$ degrees of freedom.
- The p-value is the probability of getting a $|t|$ this large or larger, if H_0 were true.
- **Rule of thumb:** $|t| \gtrsim 2$ usually gives $p < 0.05$ for moderate n .

One-Sample t-Test: Output and Interpretation

- In : `t.test(x, mu = 90)`
- Key output to report:
 - t -statistic and degrees of freedom
 - p-value
 - 95% confidence interval for μ
 - Sample mean $\bar{X}$
- **Example write-up:** “Mean fasting glucose in treated mice was 96.4 mg/dL (95% CI: 92.1–100.7), significantly higher than the reference value of 90 ($t_{24} = 2.84$, $p = 0.009$).”
- Note: always report the *effect* (how far from the reference?) alongside the p-value.

Two-Sample t-Test: When to Use It

- **Question:** Do two independent groups have different means?
- **Biological example:** You measure gene expression in tumor tissue from patients who responded to treatment ($n_1 = 20$) vs. those who did not ($n_2 = 18$). Is mean expression different between groups?
- $H_0: \mu_1 = \mu_2$ (no difference in population means)
- $H_A: \mu_1 \neq \mu_2$
- **Key requirement:** The two groups must be *independent* – different subjects, no natural pairing between observations. If measurements are paired, use the paired t-test instead (next slide).

Two-Sample t-Test: The Mechanics

- The test statistic compares the difference in sample means to the variability we would expect by chance:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{SE_{\text{diff}}}$$

- The standard error of the difference accounts for both group sizes and both group variances.
- **Equal variances?** By default, uses Welch's t-test, which does *not* assume the two groups have equal variance. This is almost always the safer choice.
- In : `t.test(x, y)` (Welch, default)
`t.test(x, y, var.equal = TRUE)` (equal variance assumed)
- When in doubt, use the default.

Two-Sample t-Test: Output and Interpretation

- Key output: t-statistic, degrees of freedom, p-value, 95% CI for $\mu_1 - \mu_2$.
- **Example write-up:** “Mean expression was 8.3 (SD = 1.4) in responders and 6.9 (SD = 1.7) in non-responders. Responders had significantly higher expression (difference = 1.4, 95% CI: 0.4–2.4; $t_{35.2} = 2.88$, $p = 0.007$).”
- **Common mistake:** reporting only the p-value without the estimated difference and its CI. The CI tells you whether the effect is scientifically meaningful.
- **Check your assumption:** Are the data approximately Normal? Use a histogram or Q-Q plot. With $n \geq 30$ per group, the CLT makes the t-test robust.

Paired t-Test: When to Use It

- **Question:** Is there a systematic difference between two measurements taken on the *same* subjects (or naturally matched pairs)?
- **Biological examples:**
 - Blood pressure measured before and after treatment in the same patients
 - Gene expression in tumor vs. adjacent normal tissue from the same patient
 - Left vs. right kidney function in the same animal
- **Why not just use a two-sample t-test?** Paired measurements are correlated – ignoring that wastes information and can give the wrong answer. The paired t-test removes between-subject variability, making it more powerful when pairing is appropriate.

Paired t-Test: The Mechanics

- Compute the *difference* for each pair: $D_i = X_{i,\text{after}} - X_{i,\text{before}}$
- Then run a one-sample t-test on the differences, testing $H_0 : \mu_D = 0$.

$$t = \frac{\bar{D}}{s_D/\sqrt{n}}$$

- This is exactly a one-sample t-test on a new variable D .
- In : `t.test(before, after, paired = TRUE)`
or equivalently: `t.test(after - before, mu = 0)`
- **Watch out:** the vectors `before` and `after` must be in the same order – row i of each must correspond to the same subject.

Paired t-Test: Output and Interpretation

- Output is identical in structure to the one-sample t-test: t , df , p -value, 95% CI for μ_D .
- **Example write-up:** “Systolic blood pressure decreased by a mean of 8.2 mmHg after treatment (95% CI: 4.7–11.7; $t_{29} = 4.77$, $p < 0.001$).”
- Note that n here is the *number of pairs*, not the total number of measurements.
- **Paired vs. two-sample:** Using a paired test when data are truly paired increases power. Using a two-sample test on paired data is not wrong per se, but it is inefficient – you are discarding structure in the data.

Chi-Square Test: When to Use It

- **Question:** Is there an association between two categorical variables?
- **Biological examples:**
 - Is disease status (case/control) associated with exposure (yes/no)?
 - Do treatment and control groups differ in the proportion of responders?
 - Is genotype (AA, Aa, aa) distributed differently across ethnic groups?
- Data are summarized in a **contingency table** of counts.
- H_0 : the two variables are independent (no association).
- H_A : the two variables are associated.

Chi-Square Test: The Contingency Table

- **Example:** 200 patients, randomized to drug vs. placebo, outcome is response (yes/no).

	Response: Yes	Response: No	Total
Drug	55	45	100
Placebo	35	65	100
Total	90	110	200

- If drug and response were *independent*, how many drug patients would we *expect* to respond? $100 \times (90/200) = 45$.
- The chi-square statistic measures the total discrepancy between observed and expected counts.

Chi-Square Test: The Mechanics

- For each cell in the table, compute:

$$\chi^2 = \sum_{\text{all cells}} \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

- A large χ^2 means the observed counts are far from what independence would predict.
- Under H_0 , χ^2 follows a chi-square distribution with $(\text{rows} - 1) \times (\text{cols} - 1)$ degrees of freedom.
- In :

```
tab <- table(group, response)
chisq.test(tab)
```

- **Assumption:** Expected counts should be ≥ 5 in each cell. If not, use Fisher's exact test: `fisher.test(tab)`.

Chi-Square Test: Output and Interpretation

- Key output: χ^2 statistic, degrees of freedom, p-value.
- **Example write-up:** “Response rates were 55% in the drug group and 35% in the placebo group. This difference was statistically significant ($\chi^2_1 = 8.08$, $p = 0.004$).”
- **Important:** The chi-square test only tells you *whether* an association exists, not *how strong* it is. Report proportions (or a relative risk / odds ratio) alongside the p-value.
- **What the chi-square test does not do:** It does not tell you *which* cell or cells are driving the association in larger tables. That requires follow-up (e.g., examining standardized residuals).

Wilcoxon Rank-Sum Test: When to Use It

- **Question:** Do two independent groups tend to have different values, when we cannot assume Normality?
- **Biological examples:**
 - Comparing survival times between two treatment arms (often skewed)
 - Comparing microbiome abundance between two groups (heavy right tail, many zeros)
 - Comparing pain scores (ordinal 1–10 scale) between groups
- Also called the **Mann-Whitney U test** – same test, different name.
- **When should you prefer this over the t-test?** When data are clearly non-Normal *and* n is small (say, $n < 30$ per group). With large n , the t-test is robust to non-Normality by the CLT.

Wilcoxon Rank-Sum Test: The Mechanics

- Instead of working with the raw values, the test works with their **ranks**.
- **Procedure:**
 - ① Pool all observations from both groups and rank them from smallest (rank 1) to largest.
 - ② Sum the ranks in group 1. Call this W .
 - ③ If the groups come from the same distribution, the rank sum W should be roughly equal between groups. A very large or small W is evidence of a difference.
- Because ranks are used instead of raw values, outliers and skewness have much less influence.
- In `wilcox.test(x, y)`
For a CI on the location shift: `wilcox.test(x, y, conf.int = TRUE)`

Wilcoxon Rank-Sum Test: Output and Interpretation

- Key output: W statistic, p-value, and (optionally) a 95% CI for the *location shift* between groups (the Hodges-Lehmann estimate).
- **Example write-up:** “Microbiome diversity scores were significantly higher in the treatment group than in controls (Wilcoxon $W = 312$, $p = 0.018$; median shift = 4.2, 95% CI: 0.8–7.6).”
- **What it tests formally:** Whether the probability that a randomly selected observation from group 1 exceeds one from group 2 is 0.5 (i.e., no systematic difference). In practice, people interpret it as a test of medians when the two distributions have roughly the same shape.
- **Paired version:** `wilcox.test(x, y, paired = TRUE)` (Wilcoxon signed-rank test).

Choosing the Right Test

Data situation	Test	Key assumption
Continuous, 1 group vs. reference	One-sample t	Approx. Normal
Continuous, 2 independent groups	Two-sample t (Welch)	Approx. Normal
Continuous, 2 paired groups	Paired t	Differences approx. Normal
Categorical, 2+ groups	Chi-square	Expected counts ≥ 5
Categorical, small samples	Fisher's exact	None
Non-Normal, 2 independent groups	Wilcoxon rank-sum	Independent observations
Non-Normal, 2 paired groups	Wilcoxon signed-rank	Paired observations

Checking Assumptions Before You Test

- **For t-tests:** Plot a histogram and a Q-Q plot of your data (or of the paired differences). Look for severe skew or outliers.
 - In : `hist(x); qqnorm(x); qqline(x)`
- **For chi-square:** Check that all expected cell counts are ≥ 5 .
 - will warn you; switch to `fisher.test()` if needed.
- **For Wilcoxon:** Observations must be independent; the test does not require Normality.
- **A common misconception:** You should not choose your test *after* looking at whether the result is significant – that inflates your false positive rate. Decide on the test based on your data type and design *before* running the analysis.

Summary

- Probability quantifies uncertainty; distributions describe how data (or a random variable) behave.
- The three workhorses for biological data: **Binomial** (counts of successes), **Poisson** (counts of rare events), **Normal** (continuous measurements).
- The **CLT** tells us that sample means are approximately Normal – this underpins most inference.
- A **p-value** measures evidence against the null hypothesis; it is not the probability the null is true.
- A **confidence interval** gives a range of plausible values; it is not a probability statement about a single interval.
- **Always pair statistical significance with effect size and biological context.**